

whereas,

$$\begin{aligned}
 e^{hA} e^{hB} &= (\text{Id} + hA + \frac{h^2 A^2}{2} + \frac{h^3 A^3}{6} + \mathcal{O}(h^4)) \\
 &\quad \times (\text{Id} + hB + \frac{h^2 B^2}{2} + \frac{h^3 B^3}{6} + \mathcal{O}(h^4)) \\
 &= \text{Id} + h(A + B) + \frac{h^2}{2}(2AB + A^2 + B^2) \\
 &\quad + \frac{h^3}{6}(A^3 + B^3 + 3AB^2 + 3A^2B) + \mathcal{O}(h^4).
 \end{aligned}$$

So the difference is

$$\begin{aligned}
 e^{hA} e^{hB} - e^{h(A+B)} &= \frac{h^2}{2}(AB - BA) \\
 &\quad + \frac{h^3}{6}(2AB^2 + 2A^2B - BA^2 - BAB - B^2A - ABA) + \mathcal{O}(h^4) \\
 &= \frac{h^2}{2}[A, B] + \mathcal{O}(h^3),
 \end{aligned}$$

where $[A, B] = AB - BA$ is the commutator of A and B .

For a splitting method, we have, replacing A by $\mathcal{L}_{H_1}$ and B by $\mathcal{L}_{H_2}$,

$$e^{h\mathcal{L}_{H_1}} e^{h\mathcal{L}_{H_2}} - e^{h\mathcal{L}_H} = \frac{h^2}{2}[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}] + \mathcal{O}(h^3).$$

Observe that, for any real-valued functions f and H of phase space,

$$\mathcal{L}_H f = \nabla f^T \mathbf{J} \nabla H = \{f, H\}.$$

Hence

$$\mathcal{L}_{H_1} \mathcal{L}_{H_2} f = \{\{f, H_2\}, H_1\},$$

and, using this and skew-symmetry of the Poisson bracket,

$$[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}]f = \{\{f, H_2\}, H_1\} - \{\{f, H_1\}, H_2\} = \{\{f, H_2\}, H_1\} + \{\{H_1, f\}, H_2\}.$$

By the Jacobi identity,

$$\{\{f, H_2\}, H_1\} + \{\{H_2, H_1\}, f\} + \{\{H_1, f\}, H_2\} = 0.$$

Therefore

$$[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}]f = -\{\{H_2, H_1\}, f\} = \{f, \{H_2, H_1\}\}.$$

This means that it is possible to relate the commutator of Lie derivatives of Hamiltonian vector fields to the Lie derivative of the Poisson bracket of the corresponding Hamiltonians, i.e.,

$$[\mathcal{L}_{H_1}, \mathcal{L}_{H_2}]f = \mathcal{L}_{\{H_1, H_2\}}f.$$

Because of this, we may interpret the first term in the expansion of the error as itself being derived from a Hamiltonian. This is a crucial observation which suggests an alternative perspective on the error expansion in splitting. The idea is to view the product of exponentials as an exponential:

$$e^{hA}e^{hB} = e^{h(A+B+hR)},$$

for some operator R . Expanding out the right hand side gives

$$\begin{aligned} e^{h(A+B+hR)} &= \text{Id} + h(A + B + hR) \\ &\quad + \frac{h^2}{2}(A + B + hR)^2 + \dots \\ &= \text{Id} + hA + hB + h^2R \\ &\quad + \frac{h^2}{2}(A^2 + AB + BA + B^2 \\ &\quad + h^2R^2 + h(A + B)R + hR(A + B)) + \dots \end{aligned}$$

Comparing this to our expansion for $e^{hA}e^{hB}$, for agreement to $\mathcal{O}(h^2)$ we must have

$$\frac{1}{2}(2AB + A^2 + B^2) = R + \frac{1}{2}(A^2 + AB + BA + B^2) = \mathcal{O}(h),$$

or

$$R = AB - \frac{1}{2}(AB + BA) = \frac{1}{2}[A, B] + \mathcal{O}(h),$$

and from this it follows that

$$e^{h\mathcal{L}_{H_1}}e^{h\mathcal{L}_{H_2}} = e^{h\mathcal{L}_G},$$

where

$$G = H_1 + H_2 + \frac{h}{2}\{H_1, H_2\} + \mathcal{O}(h^2).$$